FIITJEE

JEE(Main)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

OPEN TEST
(Main)

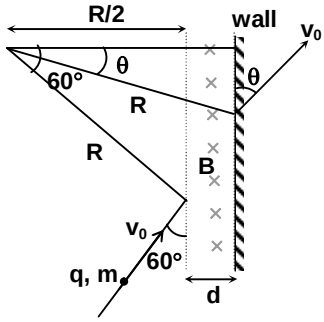
Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	**D**	31.	**C**	61.	**A**
2.	**C**	32.	**C**	62.	**C**
3.	**B**	33.	**C**	63.	**C**
4.	**B**	34.	**B**	64.	**C**
5.	**D**	35.	**C**	65.	**B**
6.	**A**	36.	**C**	66.	**B**
7.	**D**	37.	**C**	67.	**B**
8.	**C**	38.	**D**	68.	**D**
9.	**C**	39.	**D**	69.	**C**
10.	**C**	40.	**C**	70.	**D**
11.	**A**	41.	**C**	71.	**B**
12.	**B**	42.	**C**	72.	**B**
13.	**C**	43.	**B**	73.	**C**
14.	**B**	44.	**D**	74.	**C**
15.	**C**	45.	**A**	75.	**A**
16.	**D**	46.	**D**	76.	**B**
17.	**C**	47.	**C**	77.	**D**
18.	**B**	48.	**D**	78.	**A**
19.	**B**	49.	**A**	79.	**A**
20.	**B**	50.	**D**	80.	**D**
21.	**C**	51.	**B**	81.	**A**
22.	**A**	52.	**B**	82.	**A**
23.	**D**	53.	**B**	83.	**C**
24.	**D**	54.	**C**	84.	**B**
25.	**D**	55.	**B**	85.	**B**
26.	**A**	56.	**C**	86.	**B**
27.	**A**	57.	**B**	87.	**C**
28.	**A**	58.	**D**	88.	**D**
29.	**C**	59.	**C**	89.	**D**
30.	**D**	60.	**D**	90.	**B**

Physics

PART – I

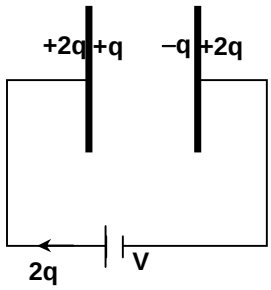
1. For the lens 'A' $\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$, v_A = 15 cm
For the lens 'B' u_2 = 10 cm, so $v_2 = \infty$
For the lens C $u_3 = \infty$ so v_3 = 30 cm

2. R cos θ – R/2 = d = $\frac{R(\sqrt{3}-1)}{2}$

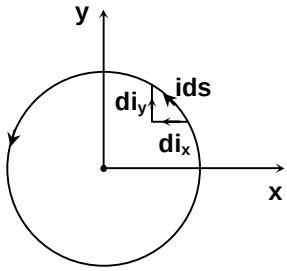


$\cos\theta-\frac{1}{2}=\frac{\sqrt{3}}{2}-\frac{1}{2}$
θ = 30°
So, total angle rotates inside the magnetic field = 30° + 30° = 60°
Time period = $\frac{m}{qB}\frac{\pi}{3}=\frac{\pi m}{3qB}$

3. The distribution of charge on the plate of capacitor.
W_B = 2qV

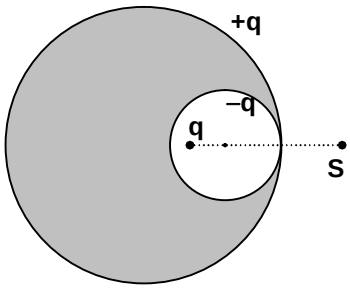


4. F_{net} = ∫ force on idy
$=4(idy)\left(\frac{B_0}{R}\right)(x)$
$=4i\frac{B_0}{R}\int_0^R\sqrt{R^2-y^2}dy$
= πiB_0R

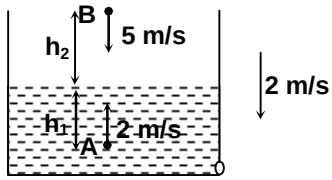


5. $\vec{E}=\frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{r}=10\hat{k}$
$\hat{r}=\hat{k}$
$\vec{B}=\frac{\mu_0}{4\pi}\frac{q\vec{v}\times\hat{r}}{r^2}=\mu_0\varepsilon_0\left\{\frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\right\}(\vec{v}\times\hat{r})$
$|\vec{B}|=\left|(10\mu_0\varepsilon_0)(\vec{v}\times\hat{r})\right|=\left|(10\mu\varepsilon_0)(4\hat{i}-3\hat{j})\right|=50\ \mu_0\varepsilon_0$

6. Electric due '–q' at 'S' is equal to electric field due to charge particle 'q'.



7. App. Position of A with respect of B.



$$P = 1\left(\frac{h_2}{1} + \frac{h_1}{\mu}\right)$$

$$P = h_2 + \frac{3}{4}h_1$$

$$V = \frac{dP}{dt} = \frac{dh_2}{dt} + \frac{3}{4}\frac{dh_1}{dt} = -3 + \frac{3}{4}(-4) = -6 \text{ m/s}$$

8. $$E = \frac{hC}{\lambda} = \frac{1242eV}{488} = 2.55 \text{ eV}$$

$$K_{max} = eV_0 = e \times 0.38 = 0.38 \text{ eV}$$

$$K_{max} = \frac{hC}{\lambda} - \phi_0 = E - \phi_0$$

$$\phi_0 = E - K_{max} = (2.55 - 0.38) = 2.17 \text{ eV}$$

9. Use basic concept

10. $$I_E = \left(\frac{100}{80}\right)I_C = \left(\frac{100}{80}\right)(10) = 12.5 \text{ mA}$$

$I_B = I_E - I_C = 12.5 - 10 = 2.5$ mA

11. The p-n junction diode acts as a half wave rectifier, so the output waveform across the resistor is positive half part.

12. Block and plank both slide

So, $a_{Block} = \frac{20-5}{2} = \frac{15}{2}$

$$a_{plank} = \frac{5}{2}$$

$$S = ut + \frac{1}{2}at^2$$

$$15 = 0 + \frac{1}{2}\frac{15}{2}t^2$$

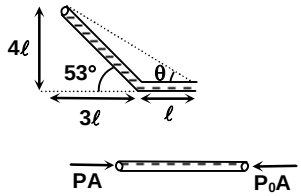
$\Rightarrow t = 2$ sec

Displacement of plane = $0 + \frac{1}{2}\frac{5}{2}2^2$

So, displacement of block with respect to plank = $15 - 5 = 10$ m

$W_f = -10 \times 20 \times 1/4 = -50$ J

13. $\tan\theta = \frac{4\ell}{4\ell} = \frac{a}{g}$



$\Rightarrow a = g$

Or, $PA - P_0A = (\rho\ell A)a$

$(5\ell)(\rho A)\left(g\frac{4}{5} - \frac{3}{5}a\right) = \rho\ell\rho a$

$a = g$

14. $V_e = \sqrt{2gR_e}$

$V_s = \frac{3}{8}\sqrt{2gR_e} = \sqrt{\frac{9gR_e}{32}} = \sqrt{\frac{R_e^2 g}{R_e + h}}$

$9R_e + 9h = 32\,R_e$

$h = \frac{23}{9}R_e$

Now, total energy at height 'h' = total energy at the surface of the earth

$0 - \frac{GM_em}{R_e} = \frac{1}{2}mv^2 - \frac{GM_em}{R_e}$

$\frac{1}{2}mv^2 = \frac{GM_em}{R_e} - \frac{GM_em}{R_e + \frac{23}{9}R_e}$

$\frac{1}{2}mv^2 = \frac{23}{32}\frac{GM_em}{R_e}$

$v = \sqrt{\frac{23}{16}\frac{GM_e}{R_e}}$, $\sqrt{gR_e} = 8$ Km/sec

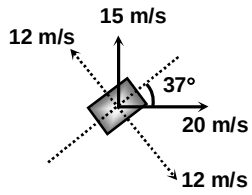
$v = \sqrt{\frac{23}{16}gR_e} = \sqrt{\frac{23}{2}}$ Km/s

15. $\ell_1 = 45$ cm, $\ell_2 = 70$ cm

$R = 100\ \Omega$

$r = R\left(\frac{\ell_1}{\ell_2 - \ell_1}\right) = 100\left(\frac{45}{75 - 45}\right) = 180\ \Omega$

16.



17. $a = v\frac{dv}{dS} = 4 + 3v$

$\int_0^v \frac{vdv}{4 + 3v} = \int_0^S dS$

$S = 1 - \frac{4}{9}\ell\left(\frac{13}{4}\right)$

18. $v_2 = \frac{60}{100}v_1 = \frac{3}{5}v_1$

$$\frac{A_t}{A_i} = \frac{2v_2}{v_1 + v_2} = \frac{2(v_2/v_1)}{1+\left(\frac{v_2}{v_1}\right)} = \frac{2(3/5)}{1+(3/5)} = \frac{3}{4}$$

19. At t = 0, x = 0

$\frac{A}{2} = A\sin\phi \Rightarrow \phi = \frac{\pi}{6}, \frac{5\pi}{6}$

Since, $\frac{dy}{dt}$ is positive for $\phi = \frac{5\pi}{6}$

20. In equilibrium extension in the string is $x_0 = \frac{mg}{k}$. For the string will remain taut in the motion maximum = $x_0 = \frac{mg}{k}$.

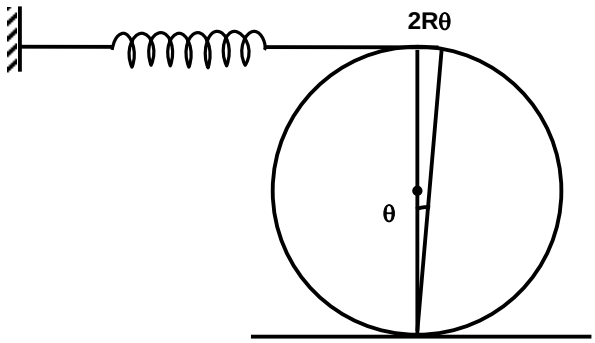
21. $\tau_P = k(2R\theta)\,2R$

$I_P\alpha = -4kR^2\theta$

$\frac{7}{5}mR^2\alpha = -4kR^2\theta$

$\alpha = -\left(\frac{20}{7}\frac{k}{m}\right)\theta$

$T = 2\pi\sqrt{\frac{7m}{20k}}$



22. $2m_1 = 3m_2$

$P_1V = \frac{m_1}{M}RT$ and $P_2V = \frac{m_2}{M}RT = \frac{2}{3}\frac{m_1}{M}RT$

$\frac{P_1}{P_2} = \frac{3}{2}$ …(i)

The process is isothermal ($P_1V_1 = P_2V_2$)

Hence, $\Delta P_1 = P_1 - \frac{P_1}{2} \Rightarrow P_1 = 2\Delta P_1$

$\Delta P_2 = P_2 - \frac{P_2}{2} = \frac{P_2}{2} = \frac{1}{2}\frac{2}{3}P_1 = \frac{P_1}{3}$

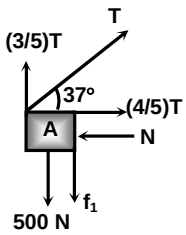
$\Delta P_2 = \frac{2}{3}\Delta P_1 = \frac{2}{3}\cdot 300$

ΔP_2 = 200 Pa

23. $C_V = \frac{3}{2}R - J/mol\cdot K = \frac{3}{2}\frac{R}{M}J/kg\cdot K = \frac{3R}{2M\times 4.2}$ cal/kgK = 0.15 Kcal/kgK

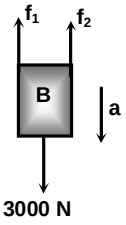
$$M = \frac{3R}{2\times 4.2\times 0.15\times 10^3} = \frac{3\times\frac{25}{3}}{2\times 4.2\times 0.15\times 10^3} = 20\times 10^{-3}\text{ kg}$$

24.



$$500 + \frac{1}{2}\frac{4}{5}T = \frac{3}{5}T \text{ N}$$

$$T = 2500$$



$$3000 - f_1 - f_2 = 300a$$

$$3000 - \frac{1}{2}\frac{4}{5}T - \frac{1}{2}\frac{4}{5}T = 3000a$$

$$3000 - \frac{4}{5}2500 = 300a$$

$$3000 - 2000 = 300\ a$$

$$a = \frac{1000}{300} = \frac{10}{3}\ \text{m/s}^2$$

25. v_1 and v_2 are the velocity after the collision.

$mv_0 \sin 37° = mv_1 \sin 37° + mv_2$ …(i)

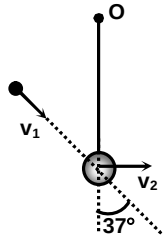
$$e = \frac{4}{5} = \frac{v_2 \cos 53° - v_1}{v_0} \quad \text{…(ii)}$$

From (i) and (ii)

$$v_2 = \frac{27}{34}v_0 = \sqrt{5g\ell}$$

$$\frac{27}{34}v_0 = 10$$

$$\Rightarrow \quad v_0 = \frac{340}{27}\ \text{m/sec}$$



26. $$\frac{1}{2}mv^2 = \mu mgx + \frac{1}{2}kx^2 \quad \text{…(i)}$$

$2\mu mg = kx$ …(ii)

From (i) and (ii)

$$v = \mu g\sqrt{\frac{8m}{k}} = 2\mu g\sqrt{\frac{2m}{k}}$$

27. $\rho_b vg = 15$ …(i)

$\rho_b vg - \rho_w vg = 12$ …(ii)

$\rho_b vg - \rho_\ell vg = 13$ …(iii)

From (i) and (iii)

$3 = \rho_w vg$

$\Rightarrow vg = 3 \times 10^{-3}$

$$\rho_b = \frac{15}{3\times 10^{-3}} = 5\times 10^3$$

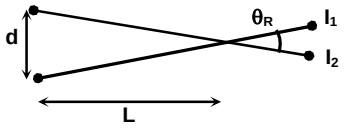
$\rho_\ell vg = 2$

$$\rho_\ell = \frac{2}{3\times 10^{-3}} = \frac{2}{3}\times 10^3$$

$$\frac{\rho_b}{\rho_\ell} = \frac{15}{2}$$

28. The work done by the external agent = change in potential energy
$= -(MB\cos\theta_2) - (-MB\cos\theta_1)$
$= -MB(\cos 60° - \cos 0°) = \frac{1}{2}MB$
$= \frac{1}{2}(1.0\times10^4\,J/T)(4\times10^{-5}T) = 0.2J$

29. $\theta_R > \frac{1.22\lambda}{d}$



$\frac{d}{L} > \frac{1.22\lambda}{d}$

$d > \frac{1.22\lambda\times L}{d}$

$> \frac{1.22\times700\times10^{-9}\times1000}{2.5\times10^{-3}}$

$> \frac{1.22\times7}{25} = 35\,cm$

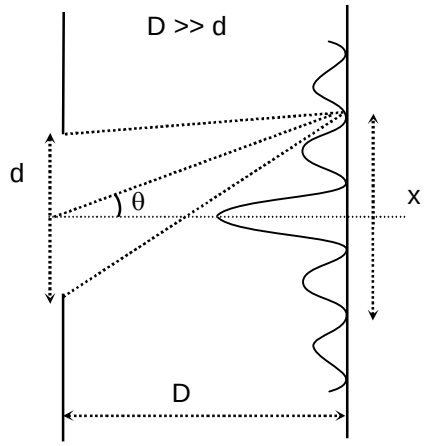
30. $\tan\theta = \frac{x}{2D}$

For second minima

$\sin\theta = \frac{2\lambda}{d} \approx \tan\theta$

$\Rightarrow \frac{x}{2D} = \frac{2\lambda}{d}$

$\Rightarrow x = \frac{4\lambda D}{d}$



Chemistry

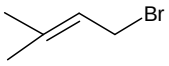
PART – II

31. Ease of NGP : O – R > C = C
Also during NGP in case of 'C' both the rings of bicyclo oxonium ion have 5 members.

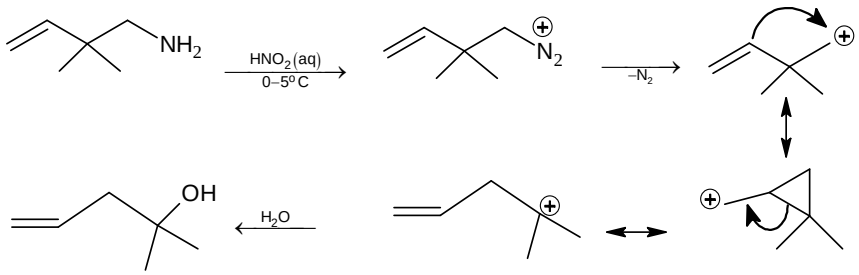
32. (A) In open system, said equilibrium is not attained.
(B) Atmospheric pressure changes with altitude.
(C) K_{eq} is the function of temperature only.
(D) K_{eq} does not give any information about the rate at which the equilibrium is reached.

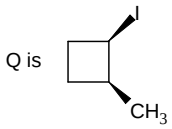
33. (A) - Chiral but optically inactive due to rapid inversion.
(B) - Achiral
(C) - Sulphur has lone pair also and no inversion occurs.
(D) - Planar and inversion occurs rapidly.

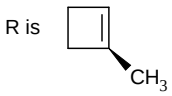
34. Reaction (I) undergoes NGP, S_N2 and S_N1; while reaction (II) follows mainly S_N1 path.

35. (IV) Major product is 

36. $BrF_3 \rightarrow [BrF_2]^+ \ [BrF_4]^-$
$IF_5 \rightarrow [IF_4]^+ \ [IF_6]^-$

37. 

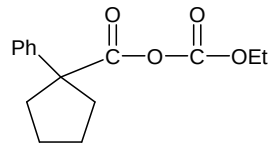
38. 

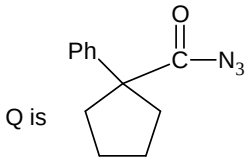


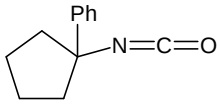
$R \xrightarrow[(2)\,H_2O_2/OH^-]{(1)\,B_2H_6/THF}$ syn addition occurs.

39. (I) Migratory aptitude of 1° alkyl > methyl group when carbocation is formed.
(II) Free carbocation formation is unfavourable in the given condition and in S, angle strain also increases much.
(III) Free carbocation is not formed under given conditions and departure of H_2O is assisted by phenyl group.

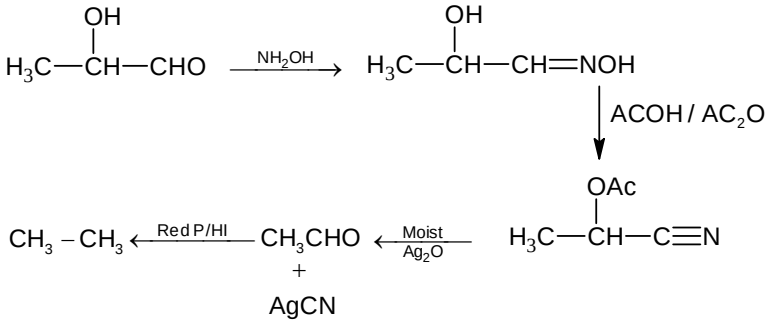
40. Bromine water can not oxidize fructose.

41. P is 

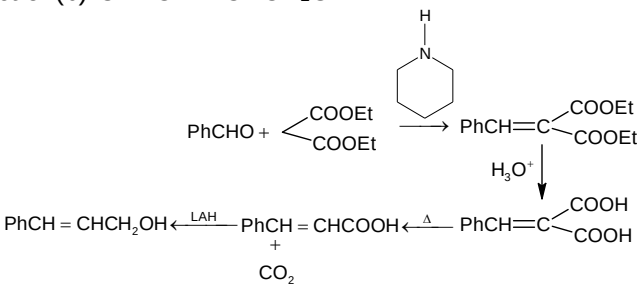
Q is 

R is 

42. Products of reactions I, II & III is propane while that of reaction IV is ethane.



43. Product of (a) and (c) is PhCH = CH – CHO;
Product of (b) is $PhCH = CHCH_2OH$
(b)



(c) $PhCHO \xrightarrow[\Delta(5-6\ hr)]{Ac_2O/AcOH} PhCH = CHCOOH \xrightarrow{LAH} PhCH = CHCH_2OH \xrightarrow{MnO_2} PhCH = CH - C$

44. $\overset{\oplus}{N}H_3 - CH_2 - COO^- (\text{glycine})$ is amphoteric.

Hence, pH = pI = $\dfrac{2.4 + 9.54}{2} = 5.97$

45. If $P_1 > P_2$; E_{cell} = +ve
And the cell reaction will be spontaneous.

46. $CN^- + H_2O_2 \longrightarrow HCO_3^- + NH_3;$

$Cl^- + AgNO_3 / HNO_3 \longrightarrow AgCl\downarrow;$

$CN^- + Hg_2^{2+} \longrightarrow \underset{(\text{Grey})}{Hg}\downarrow + \underset{(\text{Soluble})}{Hg(CN)_2};$

$Cl^- + Hg_2^{2+} \longrightarrow \underset{(\text{White})}{Hg_2Cl_2}\downarrow;$

$NaCN + \text{conc. } H_2SO_4 \longrightarrow CO\uparrow + (NH_4)_2SO_4 + Na_2SO_4;$

$Cl^- + \text{conc.}H_2SO_4 \longrightarrow HCl + HSO_4^-$

CO(g) burns with blue flame.

47. Number of electrons exchanged = 12 (for 3Sn)

$E^o_{cell} = 0.009 + 1.33 = 1.339 = \frac{0.0591}{12}\log K_{eq}.$

$K_{eq.} = 10^{271.878}$

$\simeq 10^{272}$

48. All three statements are basic concepts of thermodynamics and are correct, as well.

49. (B) & (C) Dulong & Petits law is better applicable to heavier elements/molecules and dependent of molecular constitutions as specific heat is dependent of the same.
(D) Specific heats of gases varies with the process of heating (constant pressure or constant volume)

50. Graph in A is plotted for endothermic reaction and graph in B is plotted for exothermic reactions actually.
(C) $E_{a_1} < E_{a_2} < E_{a_3}$ is correct.

51. Using Rault's law
$P^o_A = 1000$ mm Hg and $P^o_B = 800$ mm Hg
As 'A' is more volatile, hence (A) and (D) are incorrect. In case of 'C' $P_A = 0.3 \times 1000 = 300$ mm Hg.

52. (A) at critical temperature, the gas does not follow Boyle's law.
(C) $T_B = 1687.5$ K as $T_C = 500$ K $= \frac{8a}{27Rb}$
(D) It would be confirm correct above its inversion temperature (3375 K)

53. $Fe^{2+} + CN^- \longrightarrow \underset{\text{Yellowish brown}}{Fe(CN)_2} \downarrow \xrightarrow{CN^-(\text{excess})} \underset{\text{Pale yellow solution}\ (P)}{\left[Fe(CN)_6\right]^{4-}} \xrightarrow{Fe^{2+}} \underset{\text{Bluish white}}{Fe\left[Fe(CN)_6\right]^{2-}}$

$Fe^{3+} + CN^- \longrightarrow \underset{(\text{Brown})}{Fe(CN)_3} \downarrow \xrightarrow{\text{Excess } CN^-} \underset{(\text{Yellow solution})\ (Q)}{\left[Fe(CN)_6\right]^{3-}} \xrightarrow{Fe^{3+}} \underset{\text{Prussian blue}}{Fe_4\left[Fe(CN)_6\right]_3} \downarrow$

54. Correct orders are:
(A) $CO_2 = SO_3 < SO_2$ (B) $PCl_3F_2 = PCl_5 < PCl_2F_3$
(D) $CH_4 < CHCl_3 < CH_2Cl_2 < CH_3Cl$

55. Without defect formula of the compound would be : $AB_3C_4D_4$.
Due to defect
Missing particles per unit cell are

A	B	C	D
$\frac{1}{4}$	0	1	1

Hence the formula is $A_{\frac{3}{4}}B_3C_3D_3$ or $AB_4C_4D_4$.

56. $Sm^{2+}\left[(Xe)4F^{6}\right]$; $Eu^{2+}\left[(Xe)4F^{7}\right]$

$Yb^{2+}\left[(Xe)4F^{14}\right]$; $Ce^{2+}\left[(Xe)4F^{1}5d^{1}\right]$

57. (A) Be & Mg don't give flame test.
(C) Ionisation energy and melting point of Li are higher.
(D) Correct reducing character order is:
CsH > RbH > KH > NaH > LiH

59. $I_2 + aq.Cl_2 \longrightarrow IO_3^- + Cl^-$;

$I_2 + S_2O_3^{2-} \longrightarrow I^- + S_4O_6^{2-}$

$I_2 + NaOH \longrightarrow NaIO_3 + NaI$

$I_2 + Br^- \longrightarrow$ No reaction.

60. Devarda's alloy contains Al, Zn & Cu

$\underset{(A)}{NaOH} + Al \xrightarrow{\Delta} NaAlO_2 + \underset{(B)}{H_2(g)}$;

$\underset{(A)}{NaOH} + Zn \xrightarrow{\Delta} Na_2ZnO_2 + \underset{(B)}{H_2(g)}$;

$\underset{(A)}{NaOH} + P_4 \xrightarrow{\Delta} NaH_2PO_2 + PH_3(g)$;

$\underset{(A)}{NaOH} + NH_4Cl \xrightarrow{\Delta} \underset{(D)}{NH_3} + NaCl + H_2O$;

Mathematics

PART – III

61. $\left|\vec{a}+\vec{b}+\vec{c}\right|^2 = \left|\vec{a}\right|^2 + \left|\vec{b}\right|^2 + \left|\vec{c}\right|^2 + 2\left(\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a}\right) \geq 16$

$\Rightarrow 2\left(\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a}\right) \geq -82$

$\Rightarrow \vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a} \geq -41$

$\Rightarrow \vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a} = -41$ holds only if $\vec{a}\times\vec{b} = 0$

$\Rightarrow \left|\vec{a}\times\vec{b}\right| = 0$

62. $\cos\sqrt{2} - \cos\sqrt{3} > \sin\sqrt{2} - \sin\sqrt{3} > 0$

63. $\alpha - \alpha^2 + \alpha^3 - \alpha^4 + ... - \alpha^{18} = 1$

$\Rightarrow \left(\alpha+\frac{1}{\alpha}\right) - \left(\alpha^2+\frac{1}{\alpha^2}\right) + \left(\alpha^3+\frac{1}{\alpha^3}\right) - ..... + \left(\alpha^9+\frac{1}{\alpha^9}\right) = 1$

$\Rightarrow 2\text{Re}\left(\sum_{r=1}^{9}(-1)^{r+1}\alpha^r\right) = 1$

$\Rightarrow \text{Re}\left(\sum_{r=1}^{9}(-1)^{r+1}\alpha^r\right) = \frac{1}{2}$

64. $\text{Sum} = \sum_{r=1}^{31}\frac{2^r\left(r^2+1\right)}{r\left(r+1\right)} = \sum_{r=1}^{31}\left(\frac{2^{r+1}\cdot r}{r+1} - \frac{2^r\left(r-1\right)}{r}\right) = \frac{2^{32}\cdot 31}{32} = 2^{27}\cdot 31$

65. The triangle ABC is right-angled triangle with orthocentre at vertex B
Locus of vertex C is y = x – 2

66. $(x-1)^4 = 16$

$\Rightarrow x - 1 = 2, -2, 2i, -2i \ \left(i = \sqrt{-1}\right)$

67. Locus is a plane which is perpendicular to AB and passes through B(2, 3, 4)

68. T > 1

69. $f(x) = 24 - 4(x-2)^2 \ \forall\ x \in (0, 4)$

70. $r \in (0, 1)$

$\Rightarrow [r] = 0$

71. $\frac{5!}{2!\cdot 3!} + 2\cdot\frac{4!}{2!}\times 2! + \frac{3!}{2!}\times 2!\times 2! + \frac{3!}{2!} + 3!$

72. Radical centre is the orthocentre of ΔABC

73. Let distance be 'd' and angle subtended be θ

$\Rightarrow \tan\theta = \frac{5}{d}$ and $\tan 2\theta = \frac{30}{d}$

$\Rightarrow \tan\theta = \sqrt{\frac{2}{3}}$

$\Rightarrow d = 5\sqrt{\frac{3}{2}}$

74. Neither tautology nor contradiction

p	q	$\sim p \vee q$	$\sim p \vee \sim q$	$(\sim p \vee q) \wedge (\sim p \wedge q)$
T	T	T	F	F
T	F	F	T	F
F	T	T	T	T
F	F	T	T	T

75. $\sum (x_i - \bar{x})^2 = 250$

$\Rightarrow \bar{v} = \frac{250}{10} = 25$

$\Rightarrow \bar{v} = 5$

$\Rightarrow$ Coefficient of variation = $\frac{\bar{v}}{\bar{x}} \times 100 = 10\%$

76. $a = (1 + \sin 2\theta)^2$
$b = (1 - \sin 2\theta)^2$
$\Rightarrow ab = \cos^4 2\theta \le 1$

77. Applying m – n theorem
$(BD + DC)\cot 60^\circ = CD \cot 45^\circ - BD \cot C$
$\Rightarrow \cot C = 2 - \sqrt{3}$
$\Rightarrow C = 75^\circ$

78. Probability = $\frac{4}{6} \times \frac{1}{4} = \frac{1}{6}$

79. Required probability = $\frac{\sum_{r=50}^{100} {}^{100}C_r}{2^{100}} = \frac{1}{2} + \frac{{}^{100}C_{50}}{2^{101}}$

80. If ABCD is a square, then angle between Asymptotes is 90º
$\Rightarrow 3 + b = 0$
$\Rightarrow b = -3$
$\Rightarrow 2 \cdot \frac{b^2}{3} = 6$

81. ΔPS_1F_2 and ΔPF_1S_2 are similar

82. $f(x, y) = 0$
$\Rightarrow e^x \ln y + e^y \ln x = 0$
$\Rightarrow f(e, 1) = e$

83. Let $g(x) = f(f(x))$
$g'(x) = 0 \Rightarrow x = 1, 2$
$g''(x) \neq 0$ for $x = 1, 2$

84. $\cos\beta\pi = \frac{1}{2}$

$\Rightarrow \beta = \pm\frac{1}{3} + 2k$, $k \in I$

$\Rightarrow \lim\limits_{x\to\pi} \frac{\alpha\cos\alpha x}{2\beta\sin(\beta x)} = -\sqrt{3}$

$\Rightarrow \frac{(-1)^{\alpha}\cdot\alpha}{2\beta\frac{\sqrt{3}}{2}} = -\sqrt{3}$

$\Rightarrow \alpha = 1,\ \beta = \frac{1}{3}$ for $(\alpha\beta)$ minimum

$\Rightarrow (\alpha\beta)_{min} = \frac{1}{3}$

85. $\int \frac{\left(2x^2+3x-2\right)-\left(1+x^2-x\right)\ln\left(1+x^2-x\right)}{(x+2)^2\left(1+x^2-x\right)}\,dx = \frac{\ln\left(1+x^2-x\right)}{x+2} + C$

86. $y = 2x - 3$ bounds minimum area with $y^2 = 4x$

$\Rightarrow m = 2$

87. $\left(x+\frac{1}{x}\right)^2 - 2 = 14$

$\Rightarrow x + \frac{1}{x} = 4$

$x^5 + \frac{1}{x^5} = \left(x^3+\frac{1}{x^3}\right)\left(x^2+\frac{1}{x^2}\right) - \left(x+\frac{1}{x}\right) = 724$

88. For skew-symmetric matrix

$a_{ij} = -a_{ji}$

$\Rightarrow$ Only six positions are to be filled

$\Rightarrow$ Number of ways = 5^6

89. Let $A = \int_0^1 \left(2x^3-3x^2+3x-1\right)^{\frac{1}{5}}dx = -\int_0^1 \left(2x^3-3x^2+3x-1\right)^{\frac{1}{5}}dx$

$\Rightarrow A = 0$

90. Area bounded by the curves is a rectangle with vertices (–1, 2), (0, 3), (1, 0) and (2, 1)